

WORDLE Gym: Benchmarking Wordle Strategies

under Standard, Adversarial, and Hidden-Mode Play

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github.com/Michael-Ov0/WordleGym · wordlegym.vercel.app

The Problem

Classic Wordle gives six guesses; here we report solve depth so weak baselines can run past six instead of tying as failures. The answer is one of 2,315 five-letter English words. After each guess, tiles paint back: **green** if right, **yellow** if misplaced, and **gray** if absent. **WordleGym** benchmarks ten strategies, summarized by six representative policies across three game flavors.

Standard. The answer is fixed before you start. Feedback is honest. The game most humans play.

Evil. The host cheats. After every guess, they pick the feedback that keeps the most candidate answers alive. “The answer” is whatever’s most adversarial given history. In worst-case solve depth, this turns out to be easier than honest Standard Wordle.

Unknown. A coin flip at game-start picks Standard or Evil; you don’t know which. Each turn’s feedback is more likely under one mode than the other, so Bayes does the work — but does the right policy hedge?

Six Strategy Families

Every strategy answers the same question — “which legal guess do I play next?” — with a different scoring rule. Let A be the answer list, G the legal guess list, $C \subseteq A$ the candidates still consistent with feedback, and $B_r(C, g)$ the subset that would produce pattern r against guess $g \in G$; $p(r) = |B_r|/|C|$.

Strategy	Score used to choose the next guess
random-valid	uniform pick from C
letter-frequency	$\arg \max_g \text{freq}(g; C)$
expected-entropy	$\arg \max_g \sum_r p(r) \log_2(1/p(r))$
minimax	$\arg \min_g \max_r B_r $
posterior-expectimax	$\arg \min_g [qE(g) + (1 - q) T(C, g)]$, where $E(g) = \sum_r B_r ^2/ C $
evil-dp	beam-memoised DP for $D(C)$ (Evil only)

expected-entropy balances buckets; minimax guards the largest one. posterior-expectimax hedges by q ; evil-dp is the only DP-style policy reported here.

Three Recurrences

Read each recurrence as the same recipe: choose the legal guess with the smallest future cost, pay 1 turn now, and recurse only if the feedback is not all-green γ . With one candidate left, $V(\{w\}) = D(\{w\}) = 1$.

Standard — honest feedback: average over possible buckets

$$V(C) = \min_g \left[1 + \sum_{r \neq \gamma, B_r \neq \emptyset} \frac{|B_r|}{|C|} V(B_r) \right]$$

Evil — adversary: continue in the largest remaining bucket

$$D(C) = \min_g \left[1 + D(T(C, g)) \right], \quad T(C, g) = \text{largest } B_r, \text{ tied by fewer greens/yellows.}$$

Unknown — hidden mode: average using the current posterior q

$$V_U(C, q) = \min_g \left[1 + \sum_{r \neq \gamma} P(r \mid C, q, g) V_U(C_r, q_r) \right]$$

Here C_r is the candidate state after feedback r , and q_r is the Bayes-updated belief that the game is Standard. Evil follows one forced branch; Standard and Unknown average over branches. That is why D fits on a laptop: Evil reaches only $\sim 2,948$ subsets, while Standard expands over weighted partitions. We use the published $V(A)=3.421$ as the comparison target.

Benchmark Results

Strategy	Standard		Evil		Unknown	
	Mean	WC	Depth	WC	Mean	WC
expected-entropy	3.465	6	5	5	4.232	6
posterior-hybrid	3.465	6	5	5	4.237	6
posterior-expectimax	3.485	5	5	5	4.248	5
minimax	3.573	6	5	5	4.287	6
letter-frequency	3.587	8	5	5	4.294	8
random-valid	4.124	9	7	7	4.996	9

evil-dp (optimal)	—	—	4	4	—	—
Published optimum	3.421	5	4	4	—	—

Representative rows from the ten-strategy benchmark: 2,315 Standard answers, deterministic Evil, 2,316 Unknown games. Evil has one forced path per strategy, so Depth=WC. Bold = best. expected-entropy and posterior-hybrid average 1.3% above published $V(A)=3.421$.

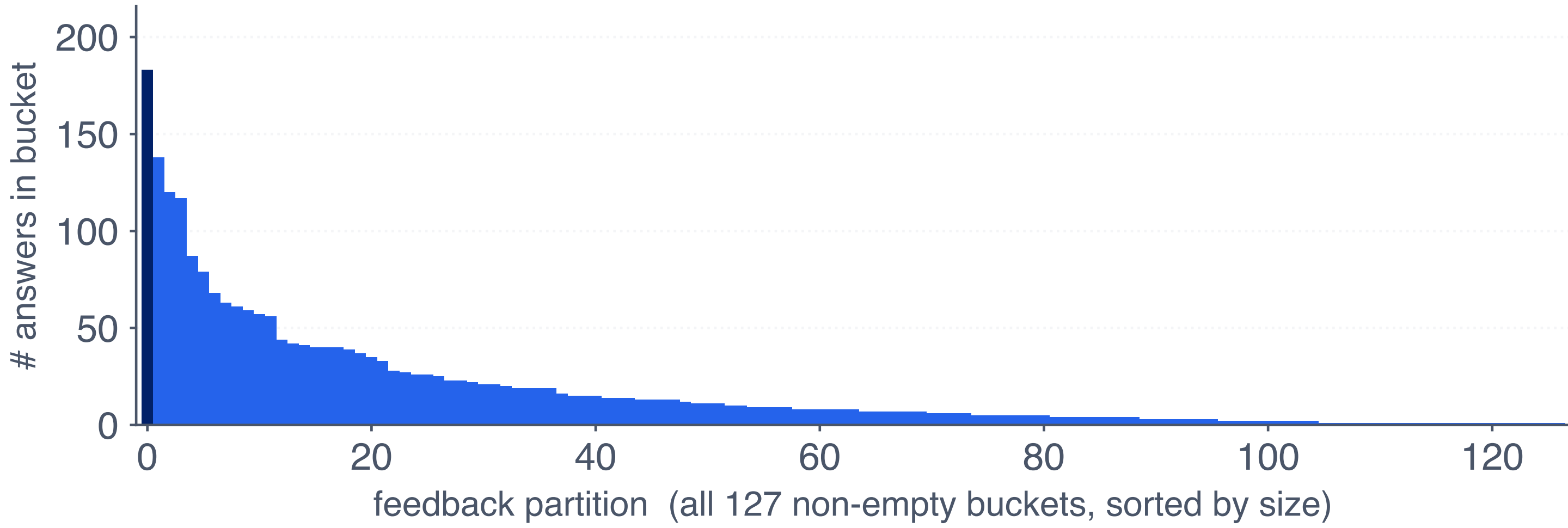
Inside the Argmin — Why the Algorithm Picks SOARE

$$H(g) = \sum_r p(r) \log_2(1/p(r)) = 5.886 \text{ bits}$$

Decision (turn 1)

S O A R E

argmax expected entropy
candidates $|C|$ **2,315**
non-empty buckets **127**
largest bucket **183**
entropy $H(g)$ **5.886 bits**



Ranking by $H(g)$

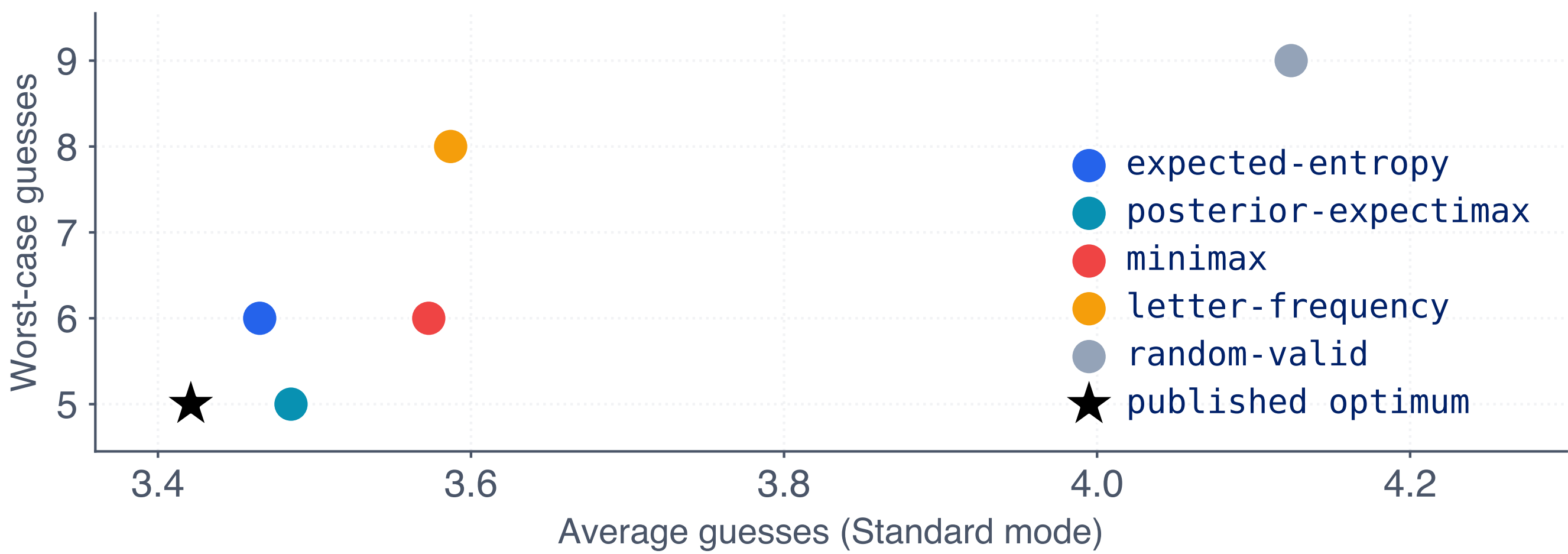
#	guess	H bits	B_max
1	SOARE	5.886	183
2	ROATE	5.883	195
3	RAISE	5.878	168
4	RAILE	5.866	173
5	SLATE	5.856	221

for contrast

19	CRANE	5.743	263
684	ADIEU	4.879	284

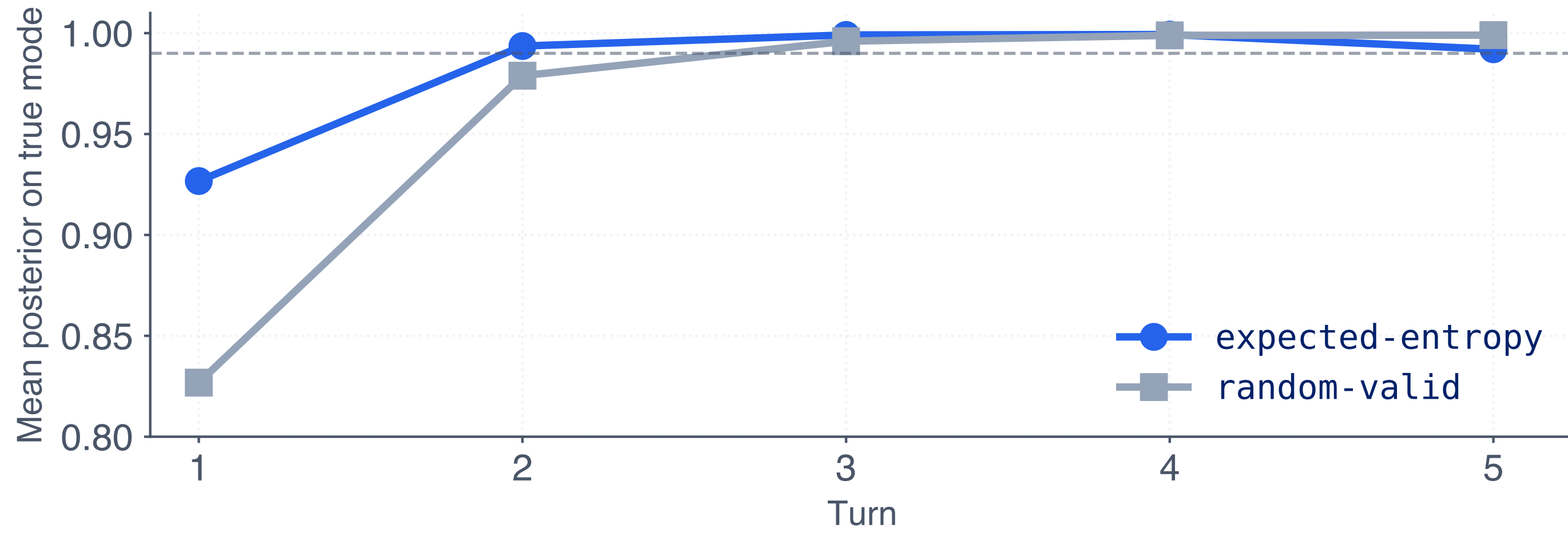
For a fixed guess, group the 2,315 answers by the feedback pattern they’d produce — the middle panel is **SOARE**’s histogram. A balanced histogram means whatever feedback comes back, you’ve eliminated about the same number of candidates — exactly Shannon’s $H(g) = \sum p(r) \log_2(1/p(r))$, which expected-entropy argmaxes. **SOARE** wins at 5.886 bits; **CRANE** has a giant bucket of 263 (worst case, barely moved); **ADIEU**’s vowel-heavy spread collapses into 80 buckets. The other strategies share the same histogram with different objectives: minimax minimises the tallest bar; letter-frequency counts shared letters; evil-dp recurses over Evil’s forced bucket.

Average vs. Worst-Case Trade-Off



Each dot is a strategy at (mean guesses, worst-case guesses) — lower-left is better, and a point dominates another if it is no worse on both axes. The headline: average and worst-case are different objectives. expected-entropy wins on average; posterior-expectimax shaves one worst-case turn for 0.02 guesses on average. The star is the published Standard optimum — the unreachable corner one-ply heuristics chase.

Why Mode-Blending Fails — the Posterior Collapses



In Unknown mode you don’t know if it’s Standard or Evil; each guess’s feedback is more likely under one, and Bayes does the rest. The y-axis is the mean posterior assigned to the *true* mode. expected-entropy crosses 0.99 between turn 1 and 2; even random-valid catches up by turn 3. Mode-aware one-ply policies hedge only while uncertain — a window that usually closes by turn 2, so posterior-expectimax’s blend has little room to help.

Findings

- One-ply lookahead is plenty.** In the table, expected-entropy averages 3.465 guesses vs. published optimum 3.421 — within 1.3%, no recursive Standard DP.
- Average and worst-case differ.** The Pareto figure shows posterior-expectimax pays +0.02 mean for one fewer worst-case turn. No free lunch.
- Evil is easier than Standard.** $D(A)=4 < W(A)=5$: the deterministic largest-bucket adversary cannot force more than four guesses; the worst honest Standard answer needs five.
- Mode-blending is a mirage.** The posterior figure shows mode probability above 99% by turn 2, so one-ply Unknown policies have little time to exploit hedging.

Code, Demo & References

github.com/wordlegym
Michael-Ov0/WordleGym .vercel.app
Supplement: wordle_gym_writeup.pdf and .tex.
Refs: Bertsimas–Paskov 2025, doi:10.1287/opre.2022.0434; Selby 2022; Sanderson 2022; Shannon 1948.